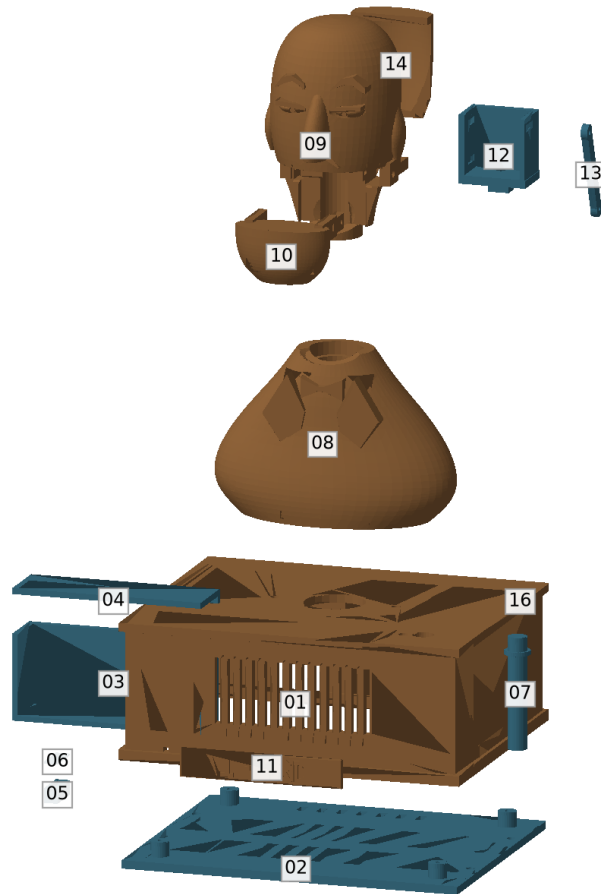


# A1-4RD / Alfred - Assembly guide

Revision A | 202 mm / 7.95 inch bust | Bambu P1 | dimensions in mm

Native SolidWorks prototype with 15 installed printed components and one fit coupon. Bronze denotes visible PLA parts; blue denotes internal PETG parts. Numbers match filenames. Exploded offsets are illustrative, not installation dimensions.



Head revision: connected 4 mm side rails, upper and lower crossmembers, reinforced axle bearings and a thicker rear neck root. The face includes solid eyes, pupils, lids and brows. Purchased electronics and fasteners are omitted from this mechanical schematic.

# Parts and print preparation

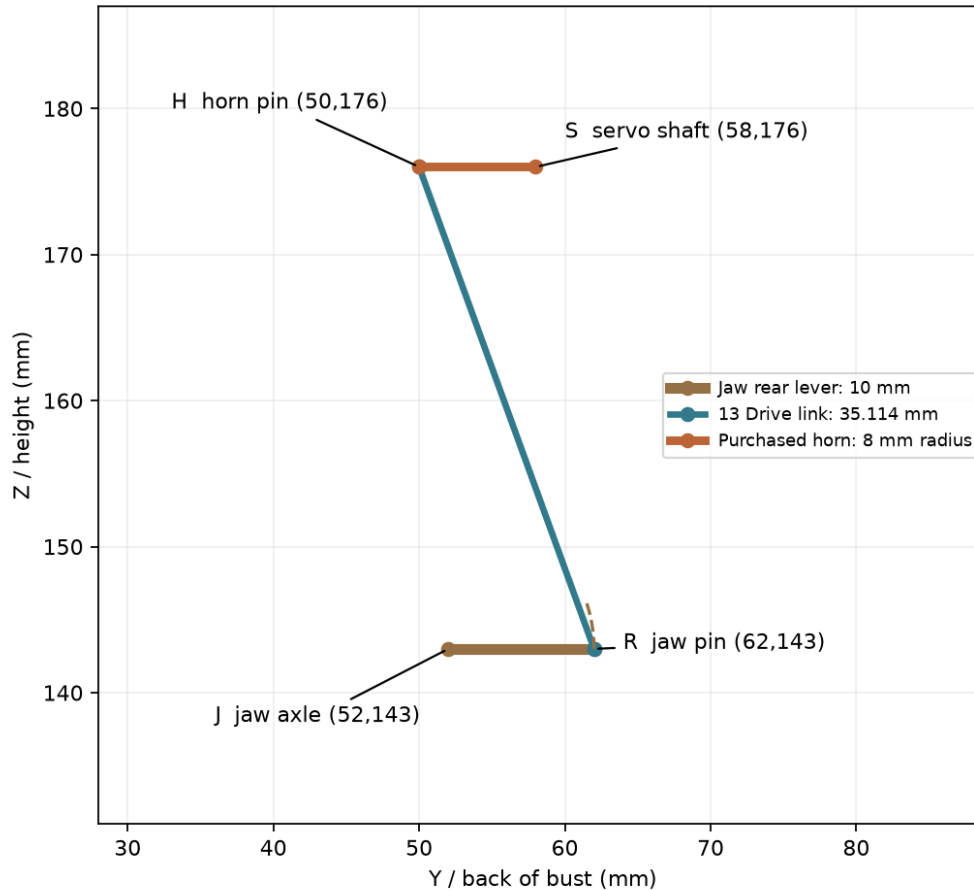
| No. | Part                | Material   |
|-----|---------------------|------------|
| 01  | Plinth              | Bronze PLA |
| 02  | Bottom cover        | PETG       |
| 03  | Speaker mount       | PETG       |
| 04  | Speaker lid         | PETG       |
| 05  | Microphone duct     | PETG       |
| 06  | Microphone duct lid | PETG       |
| 07  | Button stem         | PETG       |
| 08  | Torso               | Bronze PLA |
| 09  | Alfred head         | Bronze PLA |
| 10  | Moving jaw          | Bronze PLA |
| 11  | Name plaque         | Bronze PLA |
| 12  | Servo cradle        | PETG       |
| 13  | Drive link          | PETG       |
| 14  | Rear cover          | Bronze PLA |
| 15  | Fit coupon          | PETG       |
| 16  | Button cap          | Bronze PLA |

**1. Print the fit coupon (15) first.** Check the actual servo body and axle fit. Then print the jaw, cradle and link before committing to the cosmetic pieces. Commercial servo shaft position is provisional; verify against the next page.

**2. Slice each STL as a separate object.** Files retain assembly coordinates. Use Lay on Face and Drop to Bed. Start with 0.4 mm nozzle, 0.12-0.16 mm face layers, 0.20 mm general layers. Use six walls on the head, four to five on mechanical supports, and solid infill on the link. Print the 0.5 mm mic lid at 0.1 mm layers.

**3. Prepare hardware.** A smooth 3 mm axle, approximately 50 mm before trimming; M2 pivot screws, spacers and locknuts; narrow servo ties; M3 PCB, bottom and torso screws. Nominal screw lengths and pilot sizes are in README. Ream axle bores for free rotation. Select screw lengths by actual stack thickness.

## Mechanical schematic - right-side view



Closed-jaw reference shown. X is perpendicular to this page. Jaw axle: Y=52, Z=143. Servo shaft target: Y=58, Z=176, axis parallel to X. Link occupies X=79.5 to 81.9. Supplied servo spline and horn are required; set horn hole radius to 8 mm.

**4. Fit the servo before final assembly.** Strap the FS90 positional servo into cradle (12). Check its shaft against the reference above. Adjust cradle or linkage if needed; the commercial servo is not modeled. Attach cradle with its M2 base screw and keep wires away from the link.

**5. Assemble the jaw on the bench.** Place jaw (10) between the reinforced head bearings. Pass the smooth axle through head and jaw, trim and retain both ends. Connect link (13) to the jaw rear pin and horn with M2 screws, spacers and locknuts. Pivots must remain loose enough to rotate.

**6. Calibrate before attaching the powered horn.** Center the servo electrically, disconnect power, then install the horn at the closed reference. Move the mechanism gently by hand through its intended range before powering it. Start with a small command range, expanding only to a free 0-20 degree jaw opening.

Jaw / geometric crank angle: 0 / 180.000; 5 / 173.513; 10 / 166.483; 15 / 158.776; 20 / 150.181 degrees. These are geometry angles, not PWM values. Firmware must be calibrated to the actual servo.

# Assembly order and final checks

**7. Build the bottom module.** Bond mic duct (05) to bottom cover (02) at its modeled position. Clean and close it with lid (06). Seal the lid bore to the PCB underside microphone port with a thin gasket. Keep adhesive out of the acoustic channel. Fit the carrier PCB on its four supports, then the Pi using its PCB mounting hardware.

**8. Install the speaker.** Pad mount (03) for the nominal 70 x 31 x 17 enclosed speaker. Bond the mount to the inner front wall of plinth (01), at the supplied model position, without loading the PCB. Add lid (04), retained with removable tape or a light strap. Keep the factory speaker enclosure intact.

**9. Fit the button.** Insert stem (07) from below the plinth guide. Bond cap (16) onto the stem from above, keeping glue out of the guide. Check the switch before bonding: this prototype assumes its actuator is 4.3 mm above PCB top, with 0.2 mm resting clearance.

**10. Attach torso and route wiring.** Fasten torso (08) from inside the plinth with two M3 screws. Pass the servo cable through the torso and neck. Leave a service loop. Fit head (09) into the torso socket; lightly bond only after alignment and mechanism checks. Leave rear access open.

**11. Connect and close.** Connect speaker, mic, button and servo according to the carrier PCB electrical documentation. This guide is mechanical and does not substitute for that circuit schematic. Secure cables clear of the jaw and Pi antenna. Fit the bottom module and check rear USB/SD access before fastening its four screws.

**12. Finish the bust.** Bond A1-4RD plaque (11) to the front recess. Fit rear cover (14), using narrow removable tape around the seam for this prototype. Check that removal still gives servo access.

**13. Test assembled behavior.** Check button return, clear sound, microphone pickup and free jaw travel. Run the actual electronics and check internal temperature before extended use. Revisit support fit and fastener tension after the first operating session.

## Validation and limits

All 16 parts passed SolidWorks solid-body checks. Static checks found no interference among the 15 printed assembly components. Jaw/link samples at every 1 degree from 0 to 20 showed no intersection with the modeled head, torso, servo cradle or rear cover. These are sampled geometry checks, not a physical load, continuous-motion or purchased-component clearance certification. See included JSON reports.

This is an unbuilt prototype. Confirm actual servo shaft position, header stack, speaker envelope and switch travel before the full print. The assembly is a fixed positional reference, not a mated motion simulation. The head/torso clearance feature must be recalculated after relevant geometry changes; see README.